

## Alan Annunciator's Modbus Specification

### Modbus Specifications:

- Both ASCII and RTU modes are supported
- LRC and CRC error check included for ASCII and RTU modes respectively
- Protocol supported with RS485 serial communication
- Baud rates supported are 1200 bps, 2400 bps, 4800 bps and 9600bps.
- ASCII Mode Specifications are as follows
  - o 1 Start Bit, 7 Data Bits, 2 Stop Bits.
  - o Message frame will start with ':' followed by the message and carriage return and line feed characters as footer
  - o Data will be sent as 2 ASCII(0 to 9 and A to F) characters
  - o 1 byte LRC error check field will be included between the message and footer
- RTU Mode Specifications are as follows
  - o 1 Start Bit, 8 Data Bits, 2 Stop Bits.
  - o Message frame will start and end with 4 or 5 character delays
  - o Data will be sent as the binary values(0 to 255)
  - o 2 byte CRC error check field will be included with the message

### Annunciator's Specification:

| Data                         | Type            | Starting Address | Quantity                      | Read / write |
|------------------------------|-----------------|------------------|-------------------------------|--------------|
| NONC Status                  | Bit accessible  | 0x00000          | 12 – Standard<br>11 – DC Fail | Read & Write |
| TNT Status                   | Bit accessible  | 0x00100          | 12 – Standard<br>11 –DC Fail  | Read & Write |
| Fault Status                 | Bit accessible  | 0x10000          | 12                            | Read only    |
| Acknowledged Faults          | Bit accessible  | 0x10100          | 12                            | Read only    |
| Relay Status                 | Bit accessible  | 0x10200          | 8                             | Read only    |
| Sequence                     | Byte accessible | 0x40600          | 1                             | Read & Write |
| Keyboard                     | Byte accessible | 0x40700          | 1                             | Write only   |
| Slave ID                     | Byte accessible | 0x40800          | 1                             | Write only   |
| Baud rate                    | Byte accessible | 0x40A00          | 1                             | Write only   |
| Maximum Faults Supported     | Byte accessible | 0x40000          | 1                             | Read only    |
| Maximum NONC / TNT available | Byte accessible | 0x40100          | 1                             | Read only    |
| No of Sequences supported    | Byte accessible | 0x40200          | 1                             | Read only    |
| Slow Flash Status            | Bit accessible  | 0x10300          | 12                            | Read only    |

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### Function codes supported:

- **Read Coil Status (01):** To read status of NONC and TNT located at 0x00000 and 0x00100 respectively. Quantity of coils accessible is depends on the selection provided in the Annunicator. Reading 1 as response for NONC status states that particular selection is in NC condition. Reading 1 as response for TNT status states that particular selection is in Non Trip condition.

#### To read NONC status:

| Query:              |         | Response          |       |
|---------------------|---------|-------------------|-------|
| Slave ID            | 01      | Slave ID          | 01    |
| Function Code       | 01      | Function Code     | 01    |
| Starting Address    | 0x00000 | Byte count        | 02    |
| Quantity to be read | 0x000C  | Data Hi           | 0x5A  |
| Error Check Field   | -----   | Data Low          | 0x0F  |
|                     |         | Error Check Field | ----- |

Data is read from 0x00000 to 0x0000B and the data read is 0x5A, 0x0F.

|                 |    |    |    |    |    |    |    |    |
|-----------------|----|----|----|----|----|----|----|----|
| Fault Index:    | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  |
| Data Hi (0x5A): | 0  | 1  | 0  | 1  | 1  | 0  | 1  | 0  |
| NONC Status:    | NO | NC | NO | NC | NC | NO | NC | NO |

- **Read Input Status (02):** Fault Status, Acknowledged Fault Status, Relay Status and Slow Flash Status are considered as discrete Inputs located at 0x10000, 0x10100, 0x10200 and 0x10300 respectively. Address 0x10200 corresponds to Trip Relay status, 0x10201 corresponds to Non Trip Relay status and 0x10203 corresponds to DC Fail Relay status. Reading 1 corresponds to particular value is set.

#### To read Fault status:

| Query:              |         | Response          |       |
|---------------------|---------|-------------------|-------|
| Slave ID            | 01      | Slave ID          | 01    |
| Function Code       | 02      | Function Code     | 02    |
| Starting Address    | 0x10000 | Byte count        | 02    |
| Quantity to be read | 0x000C  | Data Hi           | 0xA5  |
| Error Check Field   | -----   | Data Low          | 0x0F  |
|                     |         | Error Check Field | ----- |

Data is read from 0x10000 to 0x1000B and the data read is 0xA5, 0x0F.

|                  |       |       |       |       |     |     |     |     |
|------------------|-------|-------|-------|-------|-----|-----|-----|-----|
| Fault Index:     | -     | -     | -     | -     | 12  | 11  | 10  | 9   |
| Data Low (0x0F): | 0     | 0     | 0     | 0     | 1   | 1   | 1   | 1   |
| Fault Status:    | Clear | Clear | Clear | Clear | Set | Set | Set | Set |

- **Read Holding Registers (03):** maximum Faults supported, Maximum NONCs /TNTs supported, No of sequences supported, Sequence Status, Keyboard, Slave ID and Baud rate are considered holding registers. Changing Slave ID and Baud rate will be effective from execution of the command and thus the response will be initialed with new slave id and new baud rate.

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### To read Sequence Status:

| Query:              |         | Response          |       |
|---------------------|---------|-------------------|-------|
| Slave ID            | 01      | Slave ID          | 01    |
| Function Code       | 03      | Function Code     | 03    |
| Starting Address    | 0x40600 | Byte count        | 02    |
| Quantity to be read | 0x0001  | Data Hi           | 0x00  |
| Error Check Field   | -----   | Data Low          | 0x03  |
|                     |         | Error Check Field | ----- |

Data is read from 0x40600 and the data read is 0x03. Address 0x40600 corresponds to Sequence Status and value read is 0x03 and means that M-Sequences selection.

| Sequence value | Sequence of Operation |
|----------------|-----------------------|
| 0x00           | F – Sequence          |
| 0x01           | S – Sequence          |
| 0x02           | N – Sequence          |
| 0x03           | M – Sequence          |

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- **Force Single Coil (05):** To modify the NONC/TNT status one at a time. To modify NONC status for fault 1 the address is 0x00000, likely for fault 2 the address is 0x00001 and so on. To send 1, data to be sent is 0xFF00 and to send 0, data to be sent is 0x0000.

### To Select NC contact for Fault 3:

| Query:            |         | Response          |         |
|-------------------|---------|-------------------|---------|
| Slave ID          | 01      | Slave ID          | 01      |
| Function Code     | 05      | Function Code     | 05      |
| Starting Address  | 0x00002 | Byte count        | 0x00002 |
| Data Hi           | 0xFF    | Data Hi           | 0xFF    |
| Data Low          | 0x00    | Data Low          | 0x00    |
| Error Check Field | -----   | Error Check Field | -----   |

Data is set at location 0x00002. Address 0x00002 corresponds to NONC Status for fault 3 and putting data as 0xFF, 0x00 sets the corresponding bit and 0x00, 0x00 clears corresponding bit.

| Data Hi, Data Low | Bit value      |
|-------------------|----------------|
| 0xFF, 0x00        | Set Bit to 1   |
| 0x00, 0x00        | Clear Bit as 0 |

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- **Preset Single Register (06):** To modify the values to a particular register, the data can be sent directly.

### To give Test:

| Query:           |         | Response      |         |
|------------------|---------|---------------|---------|
| Slave ID         | 01      | Slave ID      | 01      |
| Function Code    | 06      | Function Code | 06      |
| Starting Address | 0x40700 | Byte count    | 0x40700 |

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|                   |       |                   |       |
|-------------------|-------|-------------------|-------|
| Data Hi           | 0x00  | Data Hi           | 0x00  |
| Data Low          | 0x70  | Data Low          | 0x70  |
| Error Check Field | ----- | Error Check Field | ----- |

Data sent to the address 0x40700 is 0x70. Address 0x40700 corresponds to keyboard status and the value sent is equivalent to Test key pressed from keyboard.

|      |                 |
|------|-----------------|
| Data | Equivalent Key  |
| 0x70 | Test Key        |
| 0xB0 | Reset Key       |
| 0xD0 | Acknowledge Key |
| 0xE0 | Mute Key        |

To change Baud rate: (Table for Baud rate Values)

| Baud rate | Decimal Value | Hexadecimal Value |
|-----------|---------------|-------------------|
| 1200 bps  | 1200          | 0x4B0             |
| 2400 bps  | 2400          | 0x960             |
| 4800 bps  | 4800          | 0x12C0            |
| 9600 bps  | 9600          | 0x2580            |

### Exception Responses Supported:

Exception responses are generated under three conditions. The exception will be sent as replacing 8<sup>th</sup> bit of function code with one.

- Function code is not supported
- In valid Address is specified to either read or write data
- In valid data sent for particular address

Exception Response:

Slave ID            01  
 Function Code    0b1----- (Binary Representation -----is Actual Function code)  
 Exception Code    01  
 Error Check Field --

| Exception Code | Exception Generated   |
|----------------|-----------------------|
| 01             | Invalid Function Code |
| 02             | Invalid Address       |
| 03             | Invalid Data          |

*Note:*

*\* Changing internal setting like Slave ID or Baud rate will effective from execution of the Query and thus the response will be initialed with new Slave ID or new Baud rate.*

*\* Fresh or New annunciator will have '255' as Slave ID, 9600 bps as Baud Rate and '0' as Broadcast ID.*

*\* Abbreviations Used: NO (Normally Open), NC (Normally Closed), NT (Non Trip)*

*\* Numbering Conventions used: 0xXX (Hexadecimal), 0bXXXXXXXXX(Binary), unless otherwise mentioned number is in decimal*